

Integration Theory MATM39: Summary of Important Results

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Chapter 1

Definition 1. Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a σ -algebra on X if:

- (a) $X \in \mathcal{A}$,
- (b) if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$,
- (c) if $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\bigcup_{i=1}^\infty A_i \in \mathcal{A}$,
- (d) if $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\bigcap_{i=1}^\infty A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i , and by (c), $\bigcup_{i=1}^\infty A_i^c \in \mathcal{A}$. By (b) again, $(\bigcup_{i=1}^\infty A_i^c)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^\infty A_i = \left(\bigcup_{i=1}^\infty A_i^c \right)^c \in \mathcal{A}$$

□

Proposition 1. Let X be a set. Then the intersection of an arbitrary nonempty collection of σ -algebras on X is a σ -algebra on X .

Proof. Let $\mathcal{C}$ be a nonempty collection of σ -algebras on X , and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X , we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ -algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\{A_i\}_{i=1}^\infty \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^\infty A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^\infty A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra on X .

□

Chapter 2

Chapter 3

Chapter 5